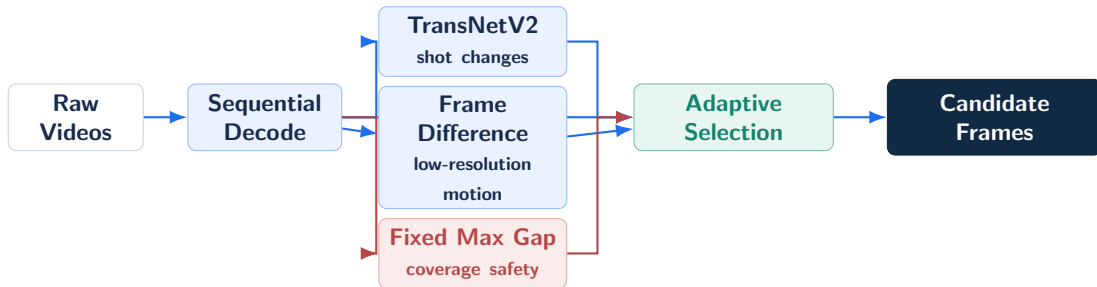


AIC 2026 Frame Preprocessing



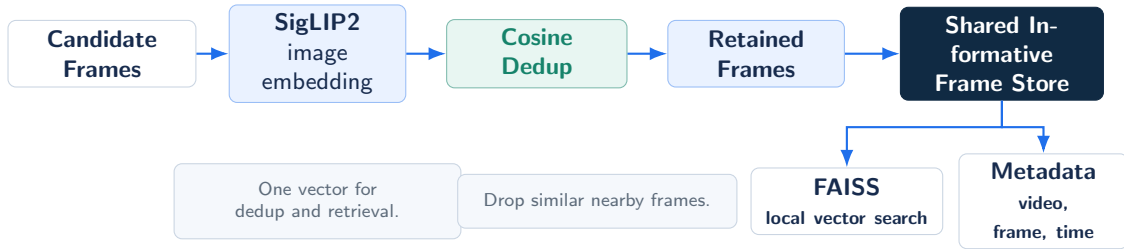
Keep fewer frames while preserving short, important events.

Offline 1 — Minimal Candidate Selection

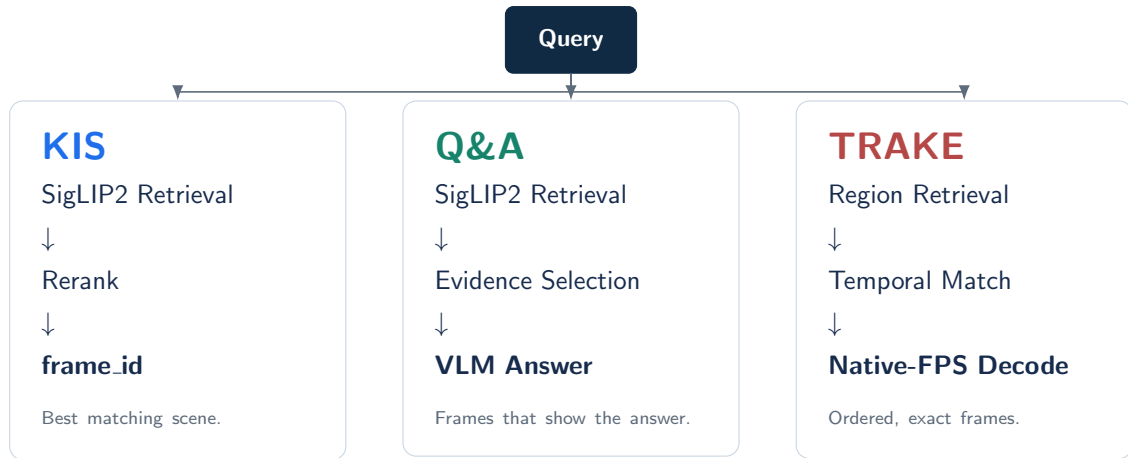


Three cheap rules select candidates; no optical-flow model is required.

Offline 2 — One Embedding Pass

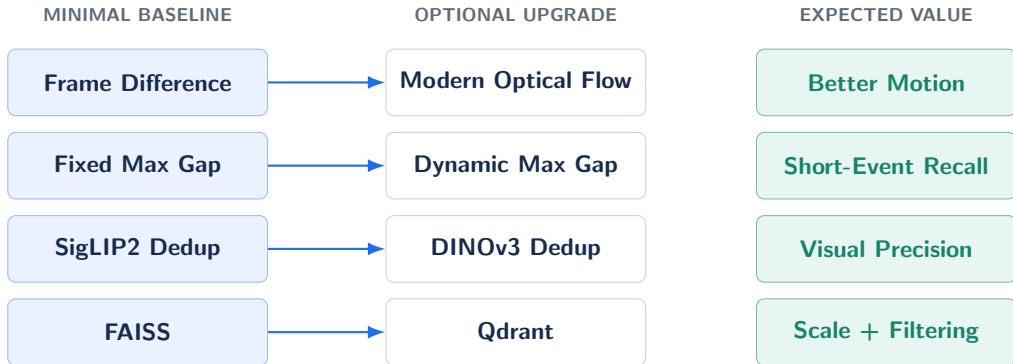


Online — Task-Specific Retrieval



One shared store; each task applies its own retrieval and reranking.

Optional Upgrades — Benchmark First



Upgrade only when AIC score or recall improves enough to justify the cost.

Minimal Pipeline, Three Clear Responsibilities

Vector Index

FAISS

Find candidate frame IDs quickly.

Shared Frame Store

Canonical Metadata

Resolve video, frame and timestamp.

Raw Video

Native-FPS Decode

Recover exact TRAKE frames locally.

Search broadly in the index – Resolve safely – Refine locally

Lower Storage

Low Latency

Preserves Short Events